

Youssef Mashaly

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EDUCATION

Georgia Institute of Technology

Graduating: **Dec 2026**

Bachelor of Science in Computer Science (Information/Internetworks and Cybersecurity)

Atlanta, GA

- **Coursework:** Data Structures & Algorithms, Computer Networking, Information Security, Cybersecurity Management, Databases, Computer Architecture, Systems & Networks, Network Security, Applied Combinatorics
- **Organizations:** Wreck Racing, Arab Student Organization

EXPERIENCE

IBM

May 2026 – Aug 2026

Software Engineering Intern (Agentic AI)

New York, NY

- Architected and built an **agentic AI** analytics platform from **9+** stakeholder discovery interviews through **system design** and implementation, enabling natural-language analysis of engineering evaluation data
- Engineered **8 modular MCP tools** with flexible argument handling, selective data retrieval, and **caching** to eliminate redundant pulls, **reduce context-window usage** and data-transfer overhead, and improve performance
- Developed **asynchronous Python** data retrieval pipelines reducing enterprise data-retrieval latency by **91.7%**
- Built release-aware **data normalization** workflows enabling cross-release analytics across evaluation datasets
- Created **planner routing** and **5 AI skills** to **orchestrate** regression, trend, delta, audit, and failure analysis

Mylo

June 2025 – Aug 2025

Software Engineering Intern

Cairo, Egypt

- Developed core features for Mylo, a leading **fintech** platform offering digital consumer finance, reducing customer credit approval time by **90%** through automating consumer onboarding processes with **Next.js** and **Amazon S3**
- Engineered an audit-logging system with **NestJS**, **PostgreSQL**, and **GraphQL**, capturing **1,000+** daily actions
- Implemented secure role-based access control (**RBAC**) using **TypeORM** and **REST APIs**, improving security
- Built reusable **React** components with configurable layouts, reducing code duplication and enhancing scalability

ACT Driving Sim Lab

Jan 2025 – Dec 2025

Undergraduate Research Assistant

Atlanta, GA

- Extended **hybrid AI driving models** using **Behavior Cloning** and **Deep Reinforcement Learning** to analyze autonomous vs. human driving behavior and improve lane-change decision-making and vehicle safety
- Optimized **PyTorch** neural-network hyperparameters, increasing lane-change prediction accuracy by **37%**
- Integrated and validated models in **CARLA** through **visual** and **numerical testing** across driving scenarios

PROJECTS

LaTeX AI Autocomplete | *JavaScript, HTML/CSS, Git*

- Boosted user LaTeX editing speed by **70%** by developing an AI-powered Google Chrome extension for Overleaf leveraging JavaScript and AI models for context-aware LaTeX command suggestions
- Integrated asynchronous **REST API** pipelines enabling context-aware suggestions from multiple AI models
- Built a dynamic interface that tracks cursor position and renders inline **context-aware** autocomplete suggestions

Spotify Wrapped | *Python, JavaScript, HTML/CSS, Git, JUnit, Django, Agile Scrum*

- Led a team of **5** to develop a full-stack web app providing **10+** interactive pages of personalized music statistics
- Implemented authentication, **CSRF** protection, internationalization, dark mode, and a **responsive UI**
- Improved runtime efficiency by **50%** by refactoring backend queries and removing unused logic

SKILLS

Languages: Java, Python, TypeScript, JavaScript, SQL, C, Bash, R, HTML/CSS, Assembly

Tools: PyTorch, React.js, Next.js, NestJS, Django, Git/GitHub, Docker, AWS, PostgreSQL, MySQL, JUnit, REST APIs, GraphQL, Hasura, Swagger/OpenAPI, OAuth, IBM watsonx Orchestrate, Langfuse, AgentOps

AI & Systems: Agentic AI, LLM Orchestration, MCP, Prompt Engineering, System Design, Async Programming

Skills: Application Development, Backend Development, API Development, Web Development, Data Processing, Cloud Computing, Low-Latency Optimization, Unit Testing, CI/CD, Test-Driven Development, Object-Oriented Programming